Transformer-Based Early Detection of Mental Health Disorders Using Multilingual Text and Emoji

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Abstract Social media helps us to understand what people think. They use a mix of language, code-switching and emojis to show how they feel. Most current methods, for detecting health issues focused on English text and ignore emojis. This limits their usefulness. In Asian communities people often use mix-language like Banglish, which is a mix of Bangla and English. They also use a lot of emojis. This makes it hard to use methods to understand their mental health. Banglish and emojis are important to understand peoples thoughts and feelings. They are a part of how people communicate in South Asia. Current methods need to be improved to consider Banglish and emojis. To address this, we present a Hybrid BERT + CNN model that incorporates multilingual BERT and RemBERT encoders along with CNN-based feature extraction and emoji preprocessing. The method can handle Bangla, English, Banglish, and text rich in emojis in a cohesive way, encompassing both the meanings and feelings of social media content. We evaluate our model using a dataset of 7,500 annotated social media posts concerning five mental health issues: Stress, Anxiety, Depression, and Suicidal Ideation. We compared our model to multiple baselines including BiLSTM, mBERT combined with BiLSTM, BiGRU with Attention, Random Forest, and SVM. Our model achieved Accuracy = 95%, Precision = 95%, Recall = 96%, and F1-Score = 94%, outperforming the other models. These findings validate that multilingual and emoji-aware transformers can be utilized to create precise and expandable systems for social media platforms.

Keywords: Mental Health Detection, Multilingual BERT , Transformer, Emoji Semantics, Social Media

1 Introduction

Social media has become a place where people share their feelings and thoughts as they happen. In informal writing many users describe their psychological

states, in a mixture of languages and using emojis. Early detection of issues like stress, anxiety, depression or suicidal tendencies is important to provide timely support. However traditional clinical approaches usually rely on self reporting or professional evaluation which can delay early identification.

In recent researches, machine learning and deep learning techniques for mental health detection based on textual data were investigated. Earlier approaches employed traditional ML models which could not capture the contextual meaning and performed poorly on social media text with noise. Advanced methods such as CNNs, LSTMs and BiLSTMs based on deep learning have improved the performance by learning textual patterns automatically. Transformer-based models like BERT and its multilingual variants have further improved performance by capturing contextual dependencies via self-attention mechanisms.

Nevertheless, some limitations still persist in this research area. Firstly, the current models tend to be mainly focused on processing English text but do not account for multilingual and code-mixed languages such as those in the South Asian context, which includes Bangla, English, and Banglish languages. Secondly, emoji use is disregarded by many researchers, but actually, emojis can be very psychologically loaded in terms of emotional cues. Thirdly, using only one model can be insufficient to capture global and local contexts.

In an attempt to solve the above-discussed problems, our system is novel in three key ways. First, unlike previous emoji-aware models [10] that deal exclusively with Bangla, our system can process Bangla, English, Banglish, emoji-laden text in a unified framework. Second, unlike code-mixed models [6] that do not consider emojis, our text processing the semantics of emojis as affective symbols that cues about emotions, in addition to text. cues simultaneously. Third, the design of two encoders utilising mBERT and RemBERT with CNN layers, allows for both broad meaning and local emotional cues such as phrases and sequences of emoji that previous research has not has not been examined in previous mental health studies. To the our best knowledge, this is the first approach to to consider all four language variants together semantics of emoji for five-class mental health mental health classification on social media from South Asia.

The main contributions of this work are as follows:

1. A pipeline which is not only multilingual but emoji aware as well and makes use of textual and symbolic emotions in its process.
2. A mixed neural model design, which employs mBERT and RemBERT alongside CNN-based feature extractors for capturing patterns at global as well as local levels.
3. A solution that works well for Bangla, English, and Banglish code-mixed social media data.
4. Extensive experimentation by comparing it to baseline models, showcasing superior results.
5. Model efficacy analysis through comparative analysis and ablation studies.

This study focuses on developing a practical approach for the early detection of mental health conditions.

2 Related Work

The emergence of social media sites has led to these platforms being recognised as valuable resources for studying human mental states. This is because individuals tend to express their thoughts more freely online compared to face-to-face interactions.

Accordingly, many researchers have utilised social media data to detect mental health conditions. Early approaches employed traditional machine learning techniques along with simple feature representations; however, their effectiveness was limited due to their inability to capture the complexity of informal text, such as misspellings, slang, and code-switching [2]. With the advancement of deep learning, models such as CNNs and LSTMs achieved improved performance by automatically learning textual patterns [11].

Subsequently, hybrid models combining recurrent neural networks and transformer-based architectures were developed to enhance contextual representation by considering the relationships between neighbouring words [9]. Significant progress has been observed with transformer-based models such as BERT and its multilingual variants, which effectively capture contextual dependencies through self-attention mechanisms [3]. Recent studies in 2025 and 2026 have increasingly focused on multilingual and code-mixed text processing, particularly in low-resource language settings, by integrating advanced transformer architectures and hybrid deep learning frameworks [1, 5].

Additionally, emojis have become an important component of communication in social media. Earlier studies often overlooked emojis, despite their strong ability to convey emotional and psychological cues. For example, emojis such as crying face and thinking face are commonly used to express emotional states in informal communication. Recent research has therefore explored emoji-aware preprocessing techniques and multimodal representations that combine textual and symbolic features to improve mental health detection performance [10].

Table 1. Comparison of Features between Existing Works and the Proposed Model

Feature	**Bokolo [4]**	**Mahajan [8]**	**Qasim [9]**	**Alakananda [1]**	**Baruah [3]**	**Proposed**
Multilingual	No	Yes	No	Yes	Yes	**Yes**
Text + Emoji	No	No	No	No	No	**Yes**
Transformer Model	Yes	No	No	Yes	Yes	**Yes**
Hybrid Architecture	No	No	Yes	Yes	Yes	**Yes**
Code-Mixed Data Handling	No	No	Partial	Yes	Yes	**Yes**
Emoji-aware Processing	No	No	No	No	No	**Yes**
Deep Learning Model	Yes	Yes	Yes	Yes	Yes	**Yes**
Real-world Social Media Data	Yes	Yes	Yes	Yes	Yes	**Yes**

3 Methodology

In this section, we introduce our proposed multilingual and emoji-sensitive system for detecting mental health disorders in Bangla, English, and Banglish social

media posts. Our model is designed to classify posts into five types: Normal, Stress, Depression, Anxiety, and Suicidal Ideation. This end-to-end detection method involves several steps: data collection, text and emoji-aware preprocessing, concurrent mBERT and emoji encoding, CNN feature extrac- ion, fusion and detection.

A core emphasis of our approach is the preservation of both textual semantics and the nuanced information conveyed by emojis, as these are vital cues in understanding mental health signals within social media communications. The total system architecture, shown in Figure 1, is specifically designed to capture and integrate these multimodal features for robust detection.

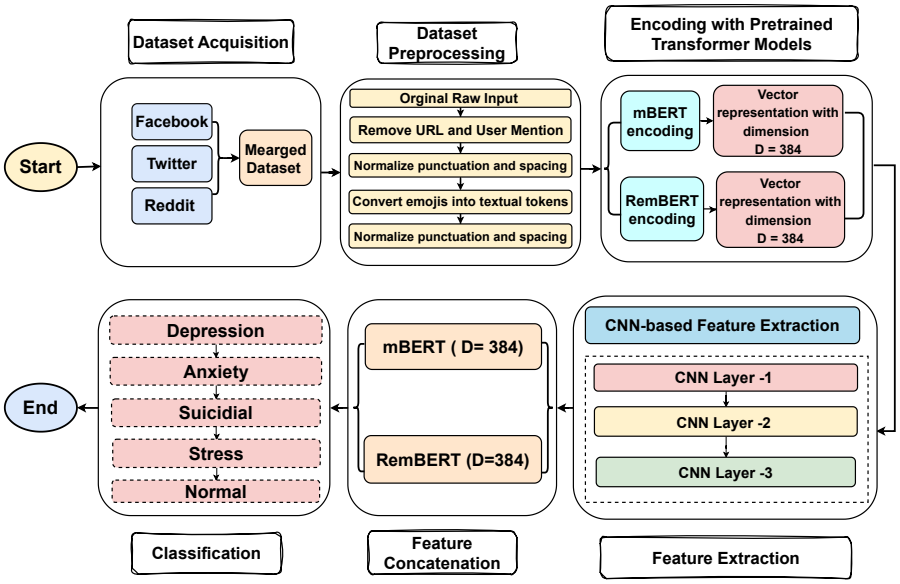

Figure 1. Proposed system architecture for mental illness classification.

3.1 Overall Architecture Description

The proposed architecture uses a hybrid deep learning approach by merging two multilingual transformer models, mBERT and RemBERT, with parallel CNN branches to extract local features. As shown in Figure 1, the pipeline takes raw social media posts as input and outputs probability scores for five mental health types.

The rationale for adopting the dual-encoder architecture is due to the capabilities of the two models. The ***mBERT*** model offers the advantage of learning a unified semantic space for Bangla, English, and mixed-language texts (Banglish). On the other hand, ***RemBERT***, being a rebalanced multilingual BERT

model, adds deep contextual learning, which is useful in analysing informal and noisy text data often encountered on social media platforms. Via encoding the same input using the pair models and then employing CNN layers to extract n-grams, the model describes global context semantics and local discriminating features, such as emotional expressions, emoji patterns and mulit-linguistic mixing, revealing mental health issues.

The whole design works as follows:

1. Data acquisition from social media platforms (Facebook, Twitter, Reddit)
2. Emoji-aware preprocessing: text cleaning, emoji demojisation, and token preparation
3. Parallel encoding using mBERT and RemBERT with linear projection to 384-dimensional representations
4. CNN-based feature extraction with multiple kernel sizes ($k \in \{2, 3, 4\}$) and global max pooling
5. Feature fusion through concatenation of the two CNN branch outputs (768-dim fused vector)
6. Fully connected classification with Softmax output over five mental health classes

The main design decision is the linear projection layers after following each transformer encoder. Since both mBERT and RemBERT produce hidden representations of dimension 768, the projection reduces each to 384 dimensions before CNN processing. The halving guarantees that the concatenated fused vector is of the same dimensionality (768) as the original transformer output, keeping the architecture consistent while injecting information from both encoders.

3.2 Data Acquisition

posts were collected from publicly available social media sites, including Facebook, Twitter, and Reddit, as these platforms frequently contain code-mixed and emoji-rich expressions related to mental states in the South Asian region. The collected posts were classified into three language groups: Bangla, English, and Banglish.

The final dataset includes of 7,500 labelled posts distributed across these five classes. Among them, 3,715 Bengali is used to write posts and 2,324 posts are written in Bengali, English. Also, the dataset contains 1,461 Banglish. Posts (Bangla-English code-mixed). Emoji usage is also occurs frequently in the dataset, where 3,749 posts have emojis. which is indicative of expressive user media behavior text.

To ensure that both textual and emoji-based information are sufficiently depicted, an annotation plan was adopted that considers both components during labelling. A voting-based annotation scheme was used to improve reliability. Initially, each post was independently labelled by multiple annotators. Subsequently, all assigned labels were compared, and the label with the majority votes was selected as the final annotation. This approach enhances the consistency and reliability of the dataset. Table 2 presents representative examples from the annotated dataset.

Table 2. Representative examples from the annotated dataset

Text	Language	Emoji	Label	Class ID
সাম্প্রতিক কাজের চাপ বেড়ে যায়; একটু বিশ্রামও পাই না 😔	Bangla	Yes	Stress	2
I'm extremely stressed with meetings every night 😟	English	Yes	Stress	2
Social situation e khub nervous lage; haat-pa kapey 😰	Banglish	Yes	Anxiety	0

3.3 Text and Emoji-Aware Preprocessing

Preprocessing is a key step in the proposed framework because social media content is often noisy, informal, and emotionally expressive. Raw posts may contain URLs, user mentions, HTML artefacts, irregular spacing, repeated punctuation, non-standard spelling, and importantly emojis that carry strong affective meaning. The preprocessing pipeline is developed to clean up textual noise. It also wants to preserve and enhance emoji-based signals.

Preprocessing Pipeline. The preprocessing pipeline is applied sequentially by removing URLs, web links, HTTP references, HTML tags, irrelevant markup symbols, and user mentions such as @username. It then normalizes punctuation, whitespace, and repeated characters while preserving emoji characters, unlike conventional NLP pipelines that often discard them. Finally, the enriched text is prepared for transformer-based tokenization.

Emoji Demojisation Strategy. A key innovation of the preprocessing stage is emoji demojisation. Each emoji is translated into its natural language description, and the generated meaning is added to the original post. This dual representation keeps both the original emoji symbol and its expanded meaning. It helps the transformer models to learn from the symbolic form and the clear meaning of emojis.

Formally, let a raw social media post be represented as:

$$S_i = \{t_i,\ e_i\} \tag{1}$$

where t_i denotes the textual component and e_i denotes the emoji component. The preprocessed input sequence is then formulated as:

$$X_i = \text{Clean}(t_i) \oplus \text{Demojize}(e_i) \tag{2}$$

where $\oplus$ denotes concatenation. For example:

Original post: *I feel hopeless* 😭💔
After preprocessing: *I feel hopeless loudly crying face broken heart*

This enriched representation ensures that emoji-based emotional cues contribute meaningfully to downstream classification, particularly for posts where the written text alone is insufficient to determine psychological state.

3.4 Parallel Transformer Encoding with mBERT and RemBERT

After preprocessing, each input sequence is independently encoded by two pre-trained multilingual transformer models **mBERT** and **RemBERT** operating in parallel on the same emoji-enriched input. This dual-encoder strategy captures complementary aspects of multilingual and informal text:

- **mBERT** provides a unified cross-lingual semantic space effective for Bangla, English, and Banglish
- **RemBERT** provides deeper rebalanced multilingual representations suited to informal social media language

Transformer Encoding Mechanism. Let the preprocessed input be represented as a token sequence $X_i = [x_1, x_2, \ldots, x_n]$. Each token is first converted to an embedding; positional encodings are then added to preserve token order:

$$\mathrm{PE}(pos,\, 2k) = \sin\Big(\frac{pos}{10000^{2k/d_{\mathrm{model}}}}\Big) \tag{3}$$

$$\mathrm{PE}(pos,\, 2k+1) = \cos\Big(\frac{pos}{10000^{2k/d_{\mathrm{model}}}}\Big) \tag{4}$$

$$E_i = X_i + \mathrm{PE}_i \tag{5}$$

The self-attention mechanism computes Query, Key, and Value matrices:

$$Q = E_i W^Q, \quad K = E_i W^K, \quad V = E_i W^V \tag{6}$$

$$\mathrm{Attention}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{7}$$

Multi-head attention enables the model to jointly attend to different representational subspaces:

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\, W^O \tag{8}$$

$$\mathrm{head}_j = \mathrm{Attention}(QW_j^Q,\ KW_j^K,\ VW_j^V) \tag{9}$$

The output of each attention layer is refined by a position-wise feed-forward network:

$$\mathrm{FFN}(x) = \mathrm{ReLU}(xW_1 + b_1)\, W_2 + b_2 \tag{10}$$

Dimensional Projection. Both mBERT and RemBERT produce contextual embeddings with hidden dimension 768:

$$H^{\mathrm{raw}}_{\mathrm{mBERT}} \in R^{T\times 768}, \qquad H^{\mathrm{raw}}_{\mathrm{RemBERT}} \in R^{T\times 768} \tag{11}$$

where T denotes the sequence length. A trainable linear projection layer reduces each representation to 384 dimensions:

$$H_{\mathrm{mBERT}} = H^{\mathrm{raw}}_{\mathrm{mBERT}}\, W_p^{\mathrm{mBERT}} + b_p^{\mathrm{mBERT}} \tag{12}$$

$$H_{\mathrm{RemBERT}} = H^{\mathrm{raw}}_{\mathrm{RemBERT}}\, W_p^{\mathrm{RemBERT}} + b_p^{\mathrm{RemBERT}} \tag{13}$$

yielding $H_{\mathrm{mBERT}} \in R^{T\times 384}$ and $H_{\mathrm{RemBERT}} \in R^{T\times 384}$. These projected embeddings encode multilingual semantic information alongside emoji-aware affective cues, and serve as inputs to the CNN extraction stage.

3.5 CNN-Based Local Feature Extraction

Transformer encoders are great at understanding the bigger picture in language by capturing relationships between words that are far apart. However, they sometimes miss shorter, local patterns like specific phrases such as "feel hopeless," "too much pressure," or even segments filled with emojis that are really important for detecting mental health issues.To solve this, CNN layers address this by performing **local n-gram feature extraction** over the transformer output sequences.

Convolutional Operation : The projected transformer output $H \in R^{T\times 384}$ is transposed to $R^{384\times T}$ and processed by one-dimensional convolutional filters with kernel size k:

$$C_k = \mathrm{ReLU}(W_k * H + b_k) \tag{14}$$

where W_k is the convolutional filter weight, b_k is the bias, and $*$ denotes convolution. Three kernel sizes are used:

$$k \in \{2,\ 3,\ 4\} \tag{15}$$

Each captures patterns of different local granularity:

- $k = 2$: short phrase-level patterns (*e.g.*, bigrams such as *"feeling sad"*)
- $k = 3$: medium-length emotional expressions (*e.g.*, *"I feel hopeless"*)
- $k = 4$: longer local semantic patterns spanning four tokens

Global Max Pooling and Feature Concatenation : Following convolution, global max pooling extracts the most discriminative :

$$P_k = \max(C_k) \tag{16}$$

The pooled outputs from all three kernel sizes are concatenated. With 128 filters per kernel size, each CNN branch produces a compact 384-dimensional feature vector:

$$O_{\mathrm{CNN}} = \mathrm{Concat}(P_2,\, P_3,\, P_4)\ \in\ R^{128\times 3} = R^{384} \tag{17}$$

Two independent CNN branches process the mBERT and RemBERT outputs respectively, yielding:

$$O_{\mathrm{mBERT-CNN}} \in R^{384}, \qquad O_{\mathrm{RemBERT-CNN}} \in R^{384} \tag{18}$$

These vectors encapsulate local emotional patterns, emoji-driven signals, and discriminative mental health phrases extracted from the transformer embeddings.

3.6 Feature Fusion

The outputs of the two CNN branches are combined via **concatenation** to form a unified fused representation:

$$F = O_{\text{mBERT-CNN}} \oplus O_{\text{RemBERT-CNN}} \in R^{384+384} = R^{768} \tag{19}$$

The resulting fused vector, which is 768 dimensions long, gives a much richer and more complete understanding than using a single encoder or just looking at the text by itself. This combined vector captures not only the words and their meanings, but also the full range of context and emotions present in multilingual mental health posts. As a result, it helps create a more thorough and nuanced picture of what someone is expressing.

3.7 Classification

The fused feature vector $F \in R^{768}$ is passed through a two-layer fully connected network with ReLU activation and dropout regularisation, followed by a Softmax output layer:

$$\hat{y} = \text{Softmax}(W \cdot F + b) \tag{20}$$

$$y_{\text{pred}} = \arg\max(\hat{y}) \tag{21}$$

where W and b are learnable parameters of the output layer. The classifier predicts one of the five mental health types:

$$\mathcal{Y} = \{\text{Normal}, \text{Stress}, \text{Depression}, \text{Anxiety}, \text{Suicidal}\}$$

Weighted cross-entropy loss is used during training to address class imbalance, ensuring that minority classes such as *Suicidal* and *Normal* receive proportionally greater gradient signal.

3.8 Training Strategy and Hyperparameter Configuration

The model is trained in two phases to balance rapid convergence with domain adaptation.

Phase 1 – Frozen Encoders : The parameters of both mBERT and RemBERT are frozen; only the projection layers, CNN branches, and classifier are updated. This allows task-specific components to converge rapidly using the pretrained transformer representations as fixed feature extractors. Training proceeds for up to 5 epochs with a learning rate of 10^{-3}, batch size 16, AdamW optimiser (weight decay 0.01), and gradient clipping at 1.0.

Phase 2 – Partial Encoder Unfreeze : The top two transformer encoder layers of each model are unfrozen and fine-tuned at a much lower learning rate of 2×10^{-5} for 2–3 additional epochs. This allows the encoders to adapt to the specific linguistic characteristics of Banglish and emoji-enriched mental health text without catastrophic forgetting of pretrained multilingual knowledge.

The chosen hyperparameters are carefully set to make sure the model trains smoothly, effective regularisation, and can be efficient fine-tuning of the proposed dual-transformer CNN architecture.Table 3 summarises the complete hyperparameter configuration.

Table 3. Hyperparameter configuration of proposed model.

Hyperparameter	**Value**	**Rationale**
MAX_LEN (sequence length)	256	Accommodates emoji demojisation expansion
Batch size	16	Balanced for dual-encoder memory footprint
CNN filters per kernel	128	As specified in the proposed architecture
CNN kernel sizes	$\{2, 3, 4\}$	Captures bi-, tri-, and 4-gram local patterns
Dropout (CNN branch)	0.2	Regularisation for local feature layers
Dropout (classifier)	0.3	Higher regularisation at decision boundary
LR — Phase 1 (frozen)	1×10^{-3}	Fast convergence for task-specific layers
LR — Phase 2 (unfrozen)	2×10^{-5}	Conservative fine-tuning to avoid forgetting
Epochs — Phase 1	5	Sufficient for CNN/classifier convergence
Epochs — Phase 2	2–3	Minimal fine-tuning of top encoder layers
Optimiser	AdamW	Weight decay = 0.01
Gradient clipping	1.0	Prevents exploding gradients
Loss function	Weighted CE	Addresses class imbalance

4 Result and Discussion

his section presents the experimental results of the proposed Hybrid BERT–CNN model for five mental health classes. The experiments were conducted on a dataset of 7,500 labeled social media posts in Bangla, English, and Banglish, with emojis preserved during preprocessing. The data were split into training, validation, and test sets using an 80%–10%–10% ratio. The proposed model is compared with five transformer based baselines, and an ablation study is conducted to evaluate the contribution of each component.

4.1 Overall Model Performance Comparison

Table 4 summarizes the performance of the proposed model and the baseline methods in terms of Precision, Recall, F1-score, and Accuracy. The proposed model outperformed the baselines by capturing the contextual semantics across Bangla, English and Banglish through multilingual transformers, while CNN layers extracted local emotional and phrase level cues. Emoji-aware preprocessing also preserves important emotive signals for mental health analysis. The comparison should be taken as relative, as the baselines are using different datasets and evaluation protocols.

Table 4. Overall model performance comparison across all metrics

Model	Precision	Recall	F1	Accuracy
Kerasiotis et al. (2024) [7]	0.90	0.88	0.89	0.88
Qasim et al. (2025) [9]	0.92	0.91	0.91	0.91
Thekkekara & Yongchareon (2025) [10]	0.93	0.92	0.92	0.92
Kawser et al. (2025) [6]	0.91	0.90	0.90	0.90
Islam et al. (2026) [5]	0.94	0.93	0.93	0.93
Proposed BERT + CNN	**0.95**	**0.96**	**0.94**	**0.95**

4.2 Ablation Study

Figure 2 presents the component-wise impact of the proposed framework. The results demonstrate performance gains from adding each module. The mBERT-only model has an accuracy of 0.90, showing that the multilingual pre-trained representations provide a strong foundation for mental health classification. The addition of RemBERT leads to a further improvement in accuracy to 0.92, which indicates that the two encoders complement each other in capturing contextual information and enhance the semantic understanding of Bangla, English, and Banglish text. Adding CNN layers to the model further increases the accuracy to 0.93. This suggests that the convolutional layers are useful in capturing local n-gram features, short emotional expressions, and other textual features that may not be widely attended to by transformer self-attention mechanisms. Finally, adding emoji-aware data preprocessing leads to another improvement, demonstrating that emojis provide useful emotional information that can be used to enhance classification decisions. The further gain shows that the semantics of emojis are effective in capturing emotional intensity and other psychological cues in social media. The complete Hybrid BERT-CNN model outperforms previous models, with an accuracy of 0.95, recall of 0.96, and F1-score of 0.94. The performance improvement across all variants shows that each of the components makes an independent contribution. Specifically, the use of two multilingual encoders, local feature extraction via CNN, and emoji-aware preprocessing better captures the information content than any of these elements alone.

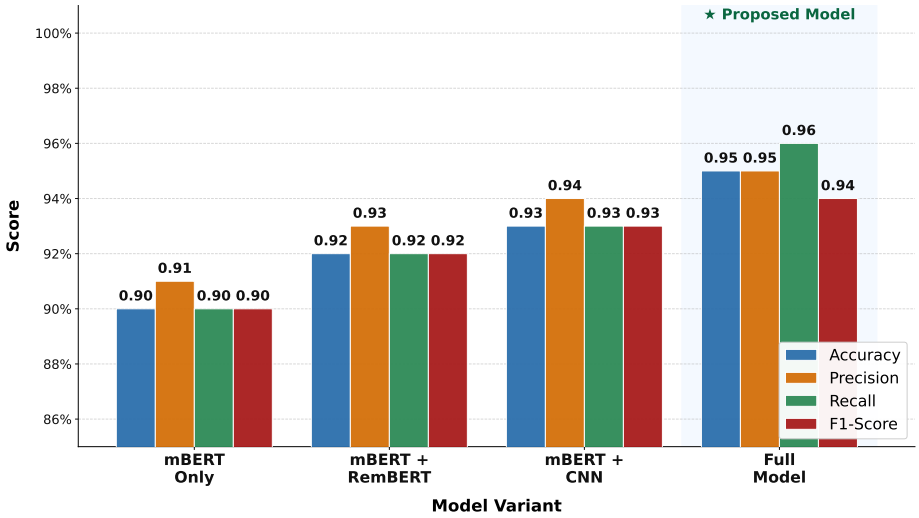

Figure 2. Ablation study: component-wise performance comparison across four model variants.

4.3 Confusion Matrix and Class-wise Risk Analysis

Figure 3 shows the confusion matrix for the 1,000 test samples. The numbers on the show that the model does a great job of classifying all five classes with about 95% of the samples being classified correctly.

The Suicidal class is the one that the model's best at identifying with 96.5% of the samples being classified correctly. This means that 193 out of 200 samples were correctly identified as Suicidal with only 5 being misclassified as Normal and 1 each as Stress and Anxiety. This is really important when it comes to detecting health issues because missing a high-risk case can have much worse consequences than a false positive. The Depression class is also identified about 95% of the time with 190 out of 200 samples being classified correctly. There is a bit of confusion between Depression and Normal and between Depression and Stress.

The biggest problem is telling the difference between Anxiety and Stress because they can look very similar in the way people express themselves on media. For example 5 samples that were actually Stress were misclassified as Anxiety and 6 samples that were actually Anxiety were misclassified as Suicidal. This shows that these categories are pretty close together. The Anxiety class is the one that the model has the trouble with with only 92.5% of the samples being classified correctly. This means that 185 out of 200 samples were classified correctly which is not surprising given how similar Anxiety and Stress can be.

The Normal and Stress classes are both identified correctly 94.5% of the time with 189 out of 200 samples being classified correctly. The mistakes that the

model makes are spread out evenly across the other categories, which suggests that there is no bias towards any particular class.

Overall the confusion matrix shows that the Hybrid BERT+CNN model does a job of classifying all five mental health categories with none of them being classified correctly less than 92.5% of the time. These results show that the model is effective at telling the difference between health conditions that are similar which is important, for detecting mental health issues. The model uses an architecture and takes into account the emojis that people use on social media, which helps it make more accurate classifications.

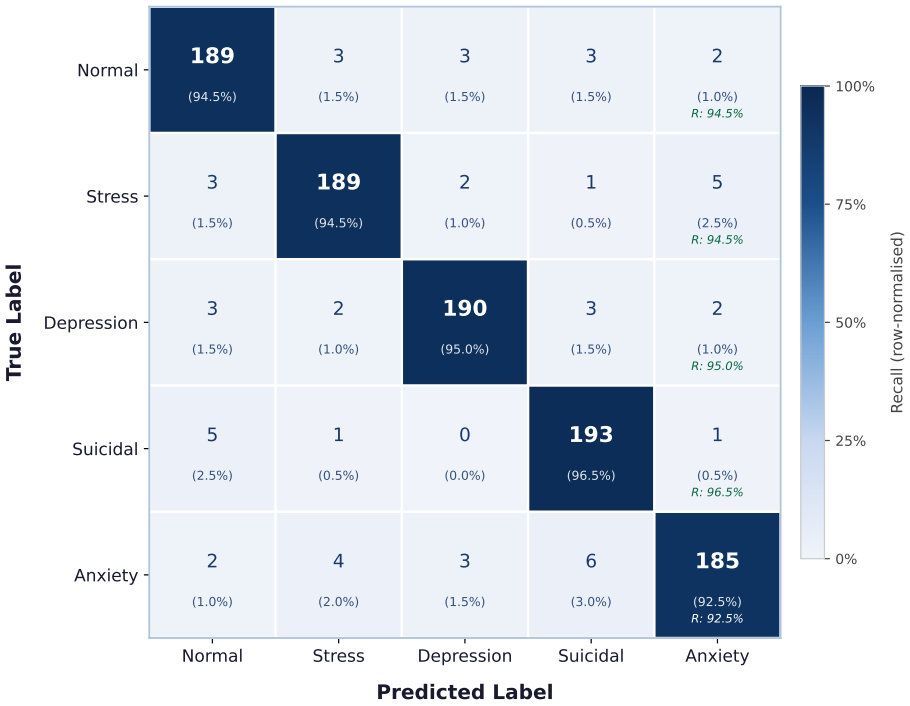

Figure 3. Confusion Matrix of the Proposed Hybrid BERT + CNN model.

4.4 Training and Validation Performance

We trained the model for 5 epochs. Train, test validation using an 80/10/10 split and early stopping (patience = 3) on the validation loss, shown in Figure 4. Validation accuracy increased steadily from 80.0% at epoch 1 to 94.0% by 5 to 94.0%, with a training accuracy of 96.0%. The steady 2% difference in training and validation accuracy during training suggests good generalisation with no overfitting.

There is a consistent decrease in the loss curves training. Training loss decreased from 0.45 to around 0.14 to about 0.19, respectively, with no gap between the two and there is no gap between the two curves in the final epochs. This shows that dropout (0.30), AdamW weight decay and class-weighting of cross-entropy loss together ensured effective regularisation and convergence over the five epochs.

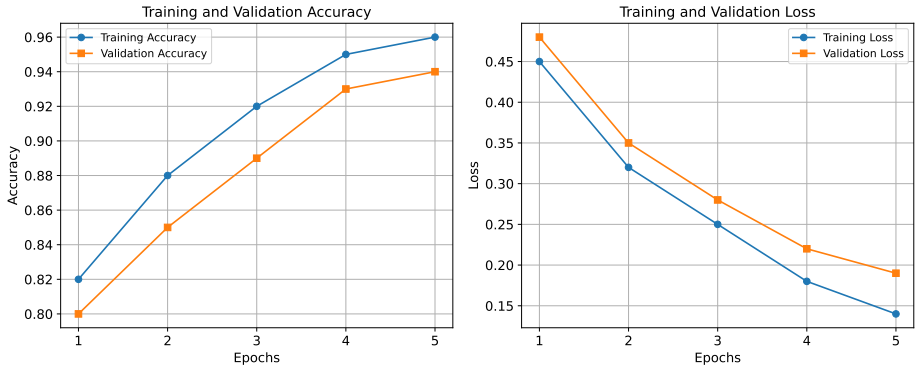

Figure 4. Training and Validation Accuracy (left) and Loss (right) over 5 epochs.

4.5 Class-wise Prediction Distribution Analysis

We collected a **7,500 social media posts** in three languages: Bengali (3,715 posts, 49.5%), English (2,324 posts, 31.0%), and Banglish (1,461 posts, 19.5%). Notably, 3,749 posts (50.0%) contain at least one emoji, which confirms that emojis are indeed widely and expressively used in social media. Table 5 shows the Precision, Recall, F1-Score, and Support of the proposed model on the test set. The findings demonstrate an overall strong and balanced performance across the five classes. The *Suicidal* class has the best Recall (0.97) metric, as missing a high-risk case is much more negative than a false positive. The *Depression* class has the highest F1-Score (0.96), which means it detects depressive language patterns. The lowest Recall is in the category *Anxiety* (0.93) because of overlap with the *Stress* and *Suicidal* categories. There is an equal number of samples (200) per class, ensuring fair evaluation. The weighted average of Precision = 0.95, Recall = 0.95, and F1 = 0.94 demonstrates that the model performs equally well across all classes.

5 Limitations

There are a number of limitations to the current study that need to be made clear to ensure replicability. First, the reported results are obtained with a

Table 5. Class-wise performance of the proposed Hybrid BERT + CNN model on the test set

Class	Precision	Recall	F1-Score	Support
Normal	0.95	0.95	0.95	200
Stress	0.96	0.95	0.95	200
Depression	0.96	0.95	0.96	200
Suicidal	0.97	0.97	0.97	200
Anxiety	0.93	0.93	0.93	200
Weighted Avg	**0.95**	**0.95**	**0.94**	**1000**

single train, validation, and test split; no statistical significance test has been conducted; therefore, the model performances with small gaps (1-2 percentage points) should be viewed with caution. Second, the proposed system uses only text and emoji but does not use other modalities like images, audio, and video, which could provide additional psychological and emotional information in social media posts. This is partly due to the extra cost and effort required to gather, label, and process multimodal data, which is a promising avenue for future research. Third, the present system is designed for offline batch processing and does not provide real-time monitoring of social media streams, which may be desirable for intervention in real-world applications. Lastly , this system has not been clinically validated and should not be used as a replacement for a professional assessment.

6 Ethical Considerations

This study used publicly available social media text for research purposes. We removed all personally identifiable information before analysis and anonymized the dataset to reduce privacy risks. We used the data solely for academic research and did not use it for surveillance, profiling, or commercial deployment. We also considered several ethical risks during model development. We examined potential bias arising from language variation and platform-specific expression patterns, and we recognized that the model could be misused if its outputs were treated as medical diagnoses. To reduce such risks, we designed the system as a research model for mental health classification rather than a replacement for professional assessment. Human oversight is required for any real-world deployment.

7 Conclusion and Future Directions

In this work we came up with a Hybrid BERT and CNN model that can detect health issues from emojis and text in many languages. We used this model to look at five kinds of health issues: Normal, Stress, Anxiety, Depression and Suicidal Ideation. Our Hybrid BERT and CNN model works with text in Bangla, English and Banglish from social media. We tried out our Hybrid BERT and

CNN model on 7,500 posts that were already labeled. It did a job: Precision was 0.95 Recall was 0.96, F1-Score was 0.94 and Accuracy was 0.95. Our Hybrid BERT and CNN model was better than recent models that use transformers. We also did some tests to see how each part of our Hybrid BERT and CNN model contributed to its success.

Several directions are identified for research on this topic. First we need to expand the dataset with dialects and more accurate annotations to improve generalisation of the model.Second it would be great to develop a real-time monitoring system for intervention on social media platforms to make a bigger impact. Third incorporating types of inputs such as images, audio and video could make the system more comprehensive and useful. Fourth we should integrate explainability tools like SHAP or LIME to help mental health professionals understand model decisions, in settings and make informed choices. Finally the model needs to undergo validation across diverse data splits before any real-world deployment of this model.

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